

Finite Element Assessment of Stress Concentrations in a Double-Lug Clevis

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1. Introduction

A lug is a structural component designed to transmit loads through a pin or bolt connection. They are widely used in mechanical and aerospace systems, where ease of assembly and reliable load transfer are required, for example in a clevis joint where structural members are connected using a pin or bolt through aligned holes. Lugs concentrate loads around the bore where the pin fits and these concentration points often lead to the onset of yielding, crack initiation or fatigue failure [1]. Empirical and photoelastic studies have shown that even under modest loads, shear and bending stresses can vary significantly depending on different design parameters [2].

Because lugs have holes and abrupt changes in geometry, they become susceptible to elevated local stress, which are also known as stress concentrations. These can arise when irregularities (such as holes or fillets) interrupt the smooth loading path through a structure which can cause the stress to crowd around this discontinuity and potentially exceed material stress limits [3]. For realistic engineering designs where safety, fatigue life or material limits are critical, only detailed analysis such as through finite element analysis (FEA) can capture these local effects sufficiently.

This study investigates how different geometric features, bore diameter and lug-base fillet radius, influence stress patterns and concentrations in a double lug clevis under pin loading. The work considers a static linear elastic load case using a 3D model of the lug in FEA. The analysis excludes modelling between the pin and lug, neglects plasticity, dynamic effects and fatigue and instead focuses on the broader elastic stress response, in order to identify optimal geometric parameters.

2. Methodology

2.1 Model Establishment

To analyse the stress distribution in a double lug clevis joint under pin loading, a 3D model was created using Abaqus CAE and CATIA V5. The baseline geometry was derived from a technical specification of an industrial double lug clevis with a 64mm bore and imported from CATIA V5 into Abaqus [4]. The geometry was validated using the distance query tool to confirm dimensions were in mm, the base geometry is shown in figure 1. The material for the clevis is assumed to be structural steel, with linear elastic isotropic properties of Young's modulus of 210,000 MPa and Poisson's ratio of 0.3 and mass density of 7850 kg/m^3 [5].

Sharp corners can produce numerical singularities in FEA models which can lead to the max stresses increasing with mesh density, preventing convergence as the force is

divided by an infinitesimally small radius. To obtain physically realistic results a small fillet (1mm) is applied to the sharp contours on the geometry, especially around the bore edge and the lug-base connection where stress gradients are highest.

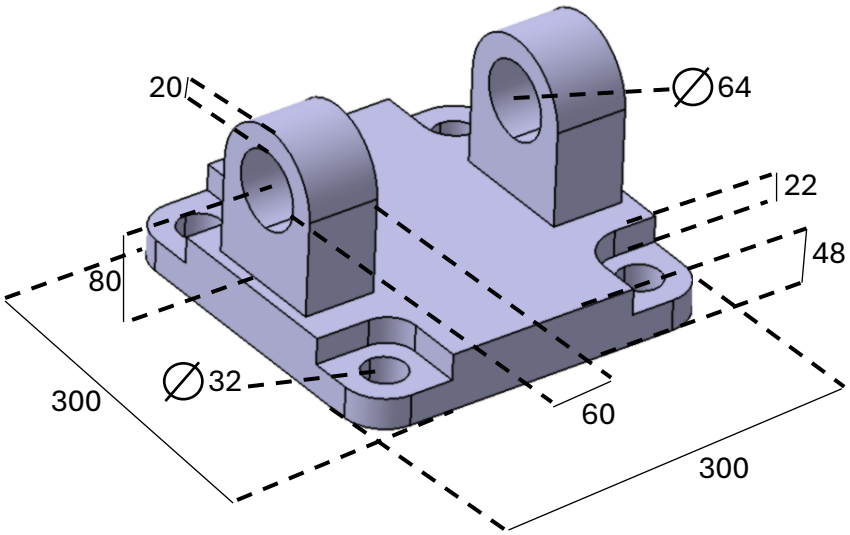


Figure 1. *Base Geometry*

2.2 Parametric Study

To explore the effects of geometry on the stress concentrations, two parameters are varied systematically whilst all other geometry parameters such as lug dimensions were kept constant, only the bore and fillet radius were modified. The study considers five different cases summarised in table 1.

The bore diameter is varied to investigate the effect of lug wall thickness and net cross-sectional area on stress and the lug-root fillet radius to understand the effect of local curvature on stress concentration.

Case	Bore Diameter, d (mm)	Fillet Radius, r (mm)
1	64	1
2	48	1
3	32	1
4	64	10
5	64	20

Table 1. *Study Parameters*

2.3 Model Simulation in Abaqus

The clevis was modelled as a 3D deformable solid and a dependent instance was created in the assembly. The pin load was simulated using a reference point (RP) positioned at the bore centre and kinematically coupled to the inner surfaces of both lugs. This assumes the pin behaves as a rigid body and distributes the load with a specified direction to the lug faces without modelling explicit contact, this can produce

slightly higher stresses as the pin cant redistribute load, however simplifies the analysis and avoids possible convergence issues [6]. As the geometry is symmetric and the lug takes the load mostly in compression around the bore, this assumption is deemed acceptable as this study focuses on overall lug stress patterns not micro scale interactions between the pin and lug.

A static linear-elastic analysis (static, general) was employed. A total vertical load of 500kN was applied through the RP, this value is slightly higher than the load capacity of a 64mm bore clevis based on a 70mm bore cited in [7], representing a conservative upper bound scenario. Linear elasticity is assumed sufficient as the stresses under this load should remain below yield while highlighting essential stress patterns and the main purpose of this study is to assess and compare the relative stress concentrations rather than predict ultimate failure.

The base of the clevis was fully fixed using a static boundary condition and constraining all degrees of freedom to prevent rigid body motion and isolate the lugs allowing the evaluation of stress and displacement directly from the pin load. The model with the boundary conditions and coupling are shown in figure 2.

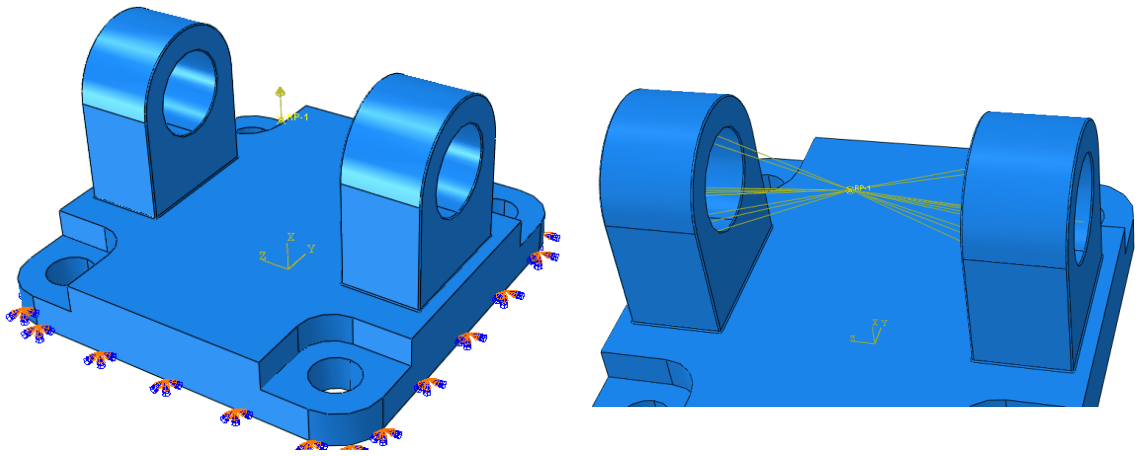


Figure 2. *Model with applied Load, Boundary Conditions and Coupling*

2.4 Meshing and Verification

The clevis was meshed using the (C3D10) quadratic tetrahedral element shape, which is a suitable option for components with complex or curved profiles [8]The automated meshing algorithm with curvature control was used to make sure the mesh automatically refined itself in regions with high curvature, such as the bore surface and lug-root fillet. This approach avoids the need for manual partitioning and seeding whilst producing a high-quality mesh capable of accurately capturing stress gradients.

Multiple different mesh densities were tested from very coarse to very fine, with element numbers spanning from ~350 to ~185,000. A convergence criterion of 3% was adopted to ensure minimal change in results between meshes whilst maintaining computational

efficiency. For each mesh the maximum displacement and stress were recorded and compared in figure 3a and 3b respectively.

Displacement stabilised by ~7000 elements where changes between consecutive mesh refinements fell below 2%. The main topic of study is the stress so it is important to prioritise its convergence, for this the maximum integration point stress and the 95th percentile stress were calculated from the Abaqus field output rather than contour legend values to avoid nodal averages, single element spikes and discrepancies caused by contour interpolation, this approach was also taken with all subsequent simulations [9], [10]. These results can be seen in table 2.

The max stress stabilises at around 220-230 MPa as the mesh is refined, the change in stress between ~67k elements and the finest mesh is ~1.8%, which is below the convergence criteria, likely converging later than displacement due to coarse meshes failing to capture stress at high gradients at the fillets. The 95th percentile value shows very little change across the finer meshes, showing in this case, the main benefit of refining the mesh is to obtain more accurate maximum stresses and stress concentrations rather than average stresses.

Therefore, for subsequent simulations the mesh sizing which produced 67,230 elements was selected for a balance of accuracy and computational cost.

Number of Elements	Max Stress (MPa)	95th Percentile Stress (MPa)
3405	112	51.6
9635	119	48.8
15993	185	49.8
29468	229	49.3
67230	226	49.6
117233	222	50.1
185154	221	49.7

Table 2. *Mesh Sensitivity Test*

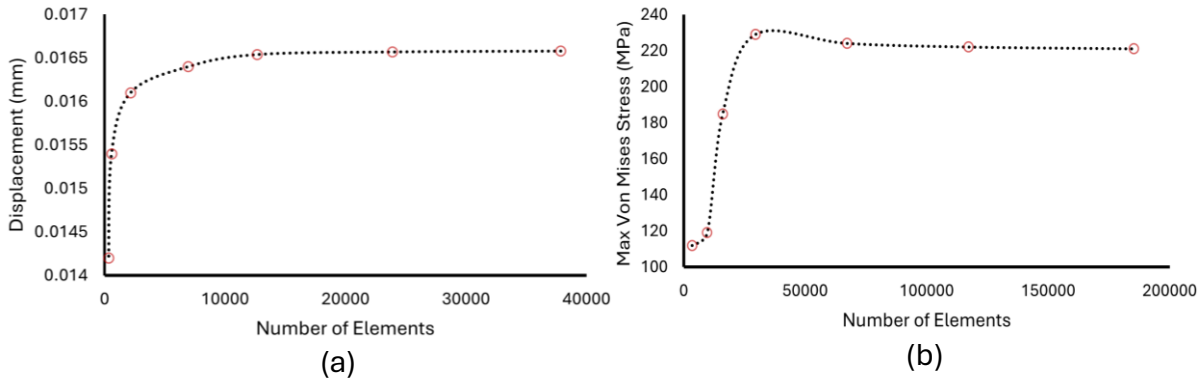


Figure 3. *Element Number vs (a) Displacement, (b) Von Mises Stress*

The nominal stress is estimated through a basic analytical method, this doesn't model the geometry in detail or account for the complexities FEA would however can serve as a reasonable baseline for validation:

$$\sigma_{nom} = \frac{F}{t * h}$$

Where $t * h$ calculates the net area through which the load is carried. F is the pin load of 250 kN (500 kN shared between the two lugs), t is the lug thickness 0.06m, h is the vertical height of the material around the bore hole $\left(0.02 + \left(\frac{0.064}{2}\right)\right) = 0.052m$. This gives a nominal stress of ~80 MPa:

$$\frac{2.5x10^5}{0.00312} \approx 8.01 \times 10^7 Pa = 80\ MPa$$

When comparing to the FEA results, the estimated nominal stress falls below the peak stress which makes sense because of stress concentrations. The 95th percentile is in the same order of magnitude which is a good sign as the nominal stress was calculated on the bore where stresses would be high but not maximal. This confirms the FEA model's general ability to capture the reasonable results. The final mesh is shown in figure 4.

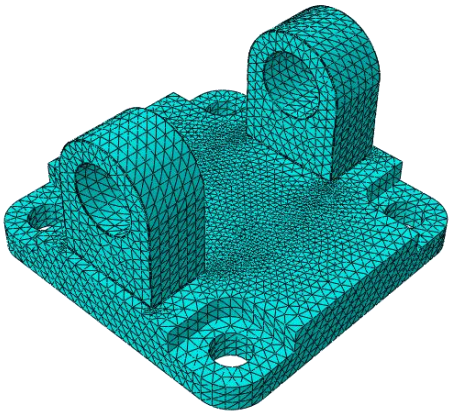


Figure 4. *Final Mesh*

3. Results and Discussion

3.1 Baseline Stress and Deformation Behavior

Before comparing the different parameter cases, the baseline configuration (d=64 mm, r=1 mm) is analysed to confirm that the model shows physically realistic stress transfer through the lugs. The displacement contour shown in figure 5a shows the entire bore region moving upwards due to the uniform load path from the kinematic coupling.

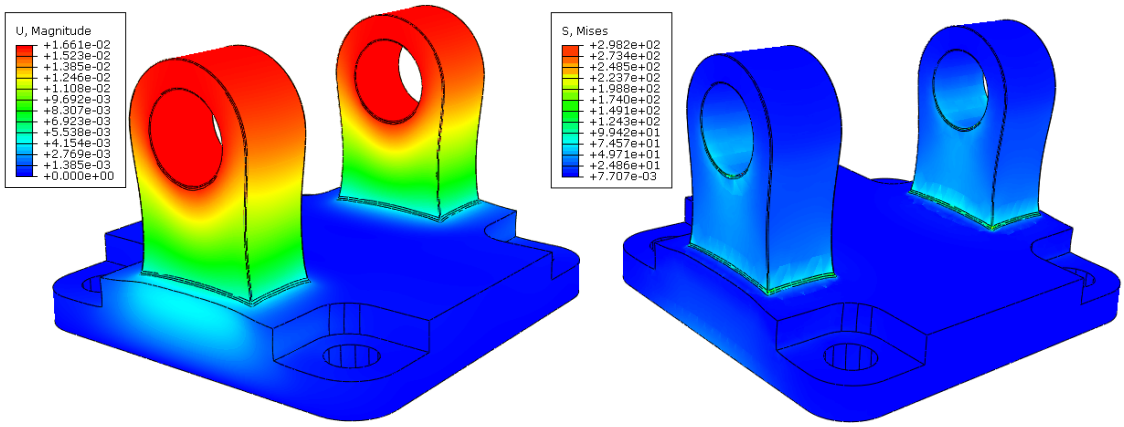


***Figure 5**. (a) Case 1 Displacement Contour, (b) Case 1 Stress Contour*

The von Mises stress contour in figure 5b shows very little stress along the top half of the bore, as the coupling simplification neglects contact that would normally create a pressure peak on the upper bore rim. As a result of this, the load is transmitted primarily through the lower ligament of the lug into the fixed base, and as expected, the highest stresses develop around the lower inside edge of the bore and at the lug-base corners due to the small fillet. A continuous band of higher stress extends from the lower bore rim through the ligament into the base, which highlights the primary load path. Both the stress and displacement contours are also symmetrical which again is expected for the symmetrical geometry and vertical load path.

This confirms the model is correctly capturing the stress and displacement patterns which are expected for the model.

3.2 Displacement Behaviour Across All Cases

The maximum vertical displacement for each case is summarised in Table 3.

Case	d (mm)	r (mm)	Displacement (mm)
1	64	1	0.0166
2	48	1	0.0190
3	32	1	0.0220
4	64	10	0.0161
5	64	20	0.0154

***Table 3.** Displacement Results*

When the bore is reduced from 64 to 48 to 32 mm, displacement steadily increases (~32% from 64 to 32 mm). A smaller bore has less contact area for load transfer, meaning the load funnels through a narrower region of material and can bend and

stretch easier under the same load. Since there is less resistance, the lug translates upwards more.

Increasing the fillet radius at the lug-base junction stiffens the load path slightly. Going from r = 1 to 10 to 20 mm reduces the displacement as a smoother lug-root transition reduces bending effects and this improved material connection improves load sharing between the bore and base. Although the reduction in displacement is small it reflects the expected trend of the fillet radius on local stiffness.

3.3 Stress Results for All Parameter Cases

The maximum integration point von Mises stress (σ_{max}) and the 95th percentile von Mises stress (σ_{95}) for each configuration are summarised in Table 4.

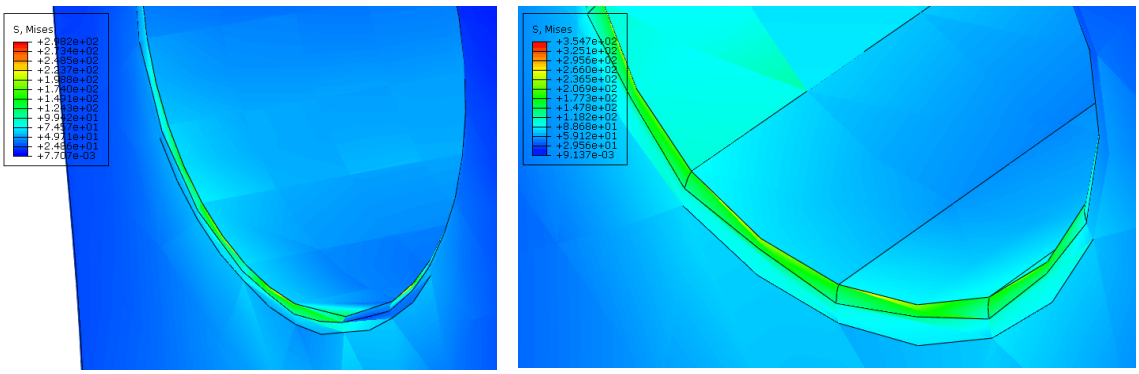
Case	d (mm)	r (mm)	σ_{max} (MPa)	σ_{95} (MPa)
1	64	1	226	49.6
2	48	1	225	52.6
3	32	1	222	56.8
4	64	10	159	48.0
5	64	20	153	46.2

Table 4. *Stress Results*

Effect of Bore Diameter on Stress

The peak stress remains within the 222-226 MPa range, essentially unchanged as the bore diameter is varied, since r is constant for cases 1 to 3 and the highest stress concentrations occur at the lug-base fillet

The overall stress level progressively increases from 49.6 to 52.5 to 56.8 MPa, this trend is expected as the smaller bores have higher bearing pressure due to the smaller load area. Figure 6 a and b show the stress around the bore hole for case 1 and case 3 respectively, from the legend, case 3 shows higher average stresses around the lower rim of the bore.



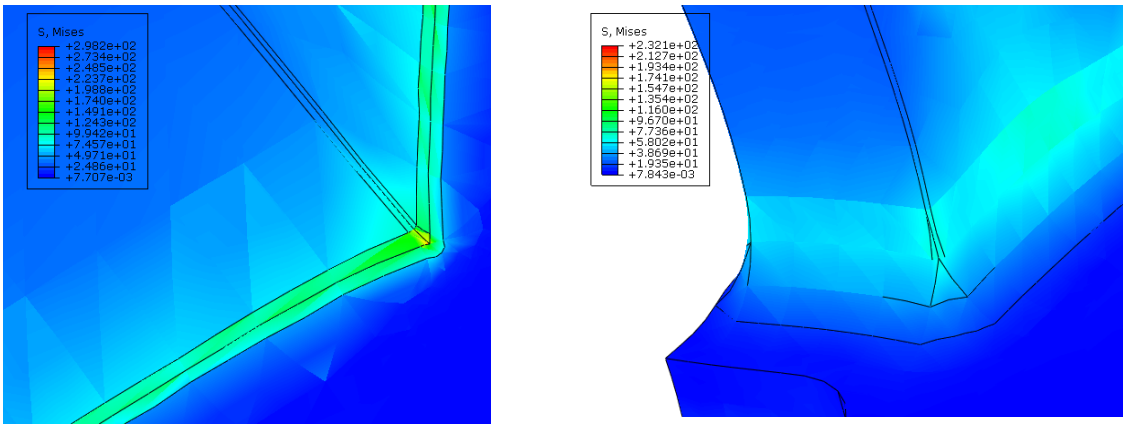
(a) (b)

Figure 6. *Stress Concentrations on Bore, (a) Case 1, (b) Case 3*

Effect of Fillet Radius on Stress Concentrations

Increasing the lug-root fillet radius (cases 1,4,5) show a significant reduction in max stress going from 226 MPa in case 1 to 153 MPa in case 5. These reductions of up to 33% are expected as a larger fillet radius distributes the stresses smoother and negates sharp curvature effects arising from extreme bending and shear in a tiny volume. σ_{95} also decreases slightly, which indicates the filleted geometry reduces not only the peak but also the average stress surrounding the lug-base region.

This effect can be represented more clearly in figure 7a and b in which the stress contour around the lug-base connection is shown for both case 1 and 5 respectively. Case 1 shows a high stress peak in the corner and higher overall stress around the connection whereas case 5 shows a far smoother transition and lower stress across the whole region.



(a) (b)

***Figure 7**. Stress Concentrations on lug-base connection, (a) Case 1, (b) Case 5*

4. Conclusion

This study evaluated how bore diameter and lug-root fillet radius influence the stress response of a double lug clevis. The model was verified through a mesh sensitivity study, which showed displacement converges faster than stresses due to steep gradients around the bore and fillet, the selected mesh (~67k elements) provided stable displacement and stress values (σ_{max} and σ_{95}).

A smaller bore diameter increases total material and reduces waste from the machining process, however, increases stress around the bore and total displacement. This highlights how despite increased total material volume there is a smaller load bearing area and this controls the stresses rather than the overall lug size. σ_{95} increased from ~49.6 to ~56.8 MPa across this range whilst displacements grew by ~35%.

Increasing the fillet radius at the lug-base showed a clear reduction in peak stress from 226 MPa at r = 1mm to 153 MPa at r = 20mm confirming that the base fillet is the dominant geometric feature governing stress concentration, making it a more powerful design parameter for extending fatigue life and lowering the risk of crack initiation. The reduction in σ_{95} and the smoothing of stress contours align with expectations indicating that fillet sizing is an effective way to manage stresses without altering bore capacity.

Several limitations remain such as the simplified load application through a kinematic coupling neglecting local pin-lug contact interactions. Material behaviour was also assumed to be linear, so plastic deformation and failure were not analysed. Including contact, nonlinear material models and cyclic loading would enable a more complete understanding of behaviour.

The study captured the governing mechanics and demonstrated clear, physically consistent trends, the results also provide a good basis for design decisions such as prioritising fillet radius to minimise stress concentrations and improve structural durability.

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